

Low Voltage Solo Mission

Name: Mohamed Mohsen

1 Competition Rules

The EV shutdown circuit contains several safety devices connected in series, including the BOTS, IMD, inertia switch and BSPD (**Rule EV6.1.2**). If the shutdown circuit opens, all AIRs and the pre-charge relay must open, accumulator current must stop immediately, and the tractive-system voltage must fall below the required safe voltage within five seconds (**Rule EV6.1.5**).

1.1 Brake System Plausibility Device (BSPD)

The BSPD detects the implausible condition of **hard braking while significant propulsion power is still being delivered**. For an EV, it must be a standalone, non-programmable circuit and open the shutdown circuit when hard braking occurs while at least 5 kW is being delivered to the motors (**Rule T11.6.1**).

The condition must persist for more than 500 ms before shutdown (**Rule T11.6.2**). Hard braking is detected using a brake pressure sensor, while EV power delivery is determined using a DC current sensor (**Rule T11.6.5–T11.6.6**).

Importance: The BSPD provides an independent hardware layer that prevents continued propulsion when the driver is trying to stop the car.

Relevant rules: T11.6.1–T11.6.10, EV6.1.5.

1.2 Insulation Monitoring Device (IMD)

The IMD continuously monitors the electrical insulation between the tractive system and the vehicle chassis.

Every EV must have an IMD (**Rule EV6.3.1**), with a response value of at least

$$500 \Omega/\text{V}$$

relative to the maximum tractive-system voltage (**Rule EV6.3.3**).

The IMD must be connected on the vehicle side of the AIRs (**Rule EV6.3.5**). If an insulation failure or IMD failure occurs, the IMD must open the shutdown circuit without programmable logic (**Rule EV6.3.7**).

A red cockpit indicator marked “IMD” must also indicate the fault (**Rule EV6.3.8**).

Importance: The IMD protects against dangerous insulation faults that could make the chassis or other conductive vehicle parts hazardous.

Relevant rules: EV6.3.1–EV6.3.8.

1.3 Accumulator Isolation Relays (AIRs)

The AIRs are the main high-voltage switching devices that connect or disconnect the accumulator from the rest of the tractive system.

At least two AIRs must be located inside each TS accumulator container (**Rule EV5.6.1**). They must disconnect both accumulator poles so that, when open, no TS voltage exists outside the accumulator container (**Rule EV5.6.2**).

The AIRs must be **mechanical, normally-open relays**; solid-state relays are prohibited (**Rule EV5.6.3**).

Their coils are powered through the shutdown circuit (**Rule EV6.1.1**). Therefore, if a safety device opens the shutdown circuit, the AIRs de-energise and physically isolate the accumulator.

Importance: The AIRs provide the final physical isolation between the stored energy in the accumulator and the external tractive system.

Relevant rules: EV5.6.1–EV5.6.3, EV6.1.1, EV6.1.5.

1.4 Brake Over-Travel Switch (BOTS)

The BOTS detects excessive brake pedal travel, which can occur if one of the hydraulic brake circuits fails.

It must be installed as part of the shutdown circuit so that brake over-travel caused by failure of at least one brake circuit opens the shutdown circuit (**Rule T6.2.1**).

Repeated operation must not re-close the shutdown circuit, and the driver must not be able to reset it (**Rule T6.2.2**).

The device must be a mechanical single-pole, single-throw switch (**Rule T6.2.3**).

Importance: If the braking system is partially damaged, the BOTS removes propulsion so that the available braking capability can be used to safely stop the vehicle.

Relevant rules: T6.2.1–T6.2.3, EV6.1.2.

1.5 Inertia Switch

The inertia switch is a mechanical crash sensor that automatically opens the shutdown circuit during a sufficiently severe impact (**Rule T11.5.1**).

It must latch after activation until manually reset.

The required trigger levels are defined in **Rule T11.5.2** as an omnidirectional peak acceleration of:

- $\leq 8g$ for a half-sine pulse of at least 50 ms,
- $\leq 13g$ for a half-sine pulse of at least 20 ms.

The device must not contain semiconductor components (**Rule T11.5.3**) and must be rigidly mounted while remaining removable for functional testing (**Rule T11.5.4**).

Importance: It automatically removes tractive power after a crash even if the driver cannot manually operate a shutdown button.

Relevant rules: T11.5.1–T11.5.4, EV6.1.2.

1.6 Tractive System Active Light (TSAL)

The TSAL provides a visible indication of the electrical state of the tractive system.

A single TSAL must be installed and used only to indicate TS status (**Rule EV4.10.1**).

The **red light** must flash at 2–5 Hz with a 50% duty cycle when the DC-link voltage exceeds the lower of:

$$60 \text{ VDC}$$

or

$$\frac{1}{2} V_{\text{TS,nominal}}$$

as defined in **Rule EV4.10.2**.

The **green light** is continuously illuminated when:

- all AIRs are mechanically open,
- the pre-charge relay is mechanically open,
- the vehicle-side TS voltage is below the defined safe limit.

These conditions are required by **Rule EV4.10.3**.

The actual mechanical relay states must be detected (**Rule EV4.10.5**), and the complete TSAL circuitry must use hard-wired electronics rather than software (**Rule EV4.10.9**).

A green cockpit indicator labelled “TS off” must also illuminate when the green TSAL condition is present (**Rule EV4.10.10**).

Importance: The TSAL allows drivers, team members, scrutineers and emergency personnel to quickly determine whether hazardous TS voltage may still be present.

Relevant rules: EV4.10.1–EV4.10.17.

1.7 Summary

Device	Detects / Controls	Safety Function
BSPD	Brake pressure and motor power	Stops propulsion during simultaneous hard braking and high motor power.
IMD	TS insulation	Shuts down the TS when insulation becomes unsafe.
AIR	Accumulator connection	Physically disconnects the accumulator from the vehicle TS.
BOTS	Brake pedal over-travel	Removes propulsion following a brake-system failure.
Inertia Switch	Crash acceleration	Automatically shuts down the TS after an impact.
TSAL	TS voltage and relay state	Shows whether dangerous TS voltage may still be present.

2 PCB and Circuits

2.1 Crosstalk in PCB Design

Crosstalk is the unwanted coupling of a signal from one PCB trace (the *aggressor*) into a nearby trace (the *victim*). It occurs because adjacent traces have parasitic capacitance and mutual inductance, allowing changing electric and magnetic fields from one signal to induce noise on another.

Crosstalk becomes more significant when:

- traces run parallel for a long distance,
- spacing between traces is small,
- signals have fast rise and fall times,
- the return-current path is poor or far from the trace,
- high-speed/noisy traces are routed close to sensitive analog or communication signals.

The main effects are unwanted voltage spikes, signal distortion, false logic transitions and increased EMI. In high-speed communication circuits, excessive crosstalk can also reduce signal integrity and increase bit errors.

Crosstalk can be reduced by:

- Increasing trace spacing
- Reducing parallel routing length
- Using a continuous reference plane
- Routing adjacent signal layers in different directions
- Separating noisy and sensitive signals
- Reducing edge rate when possible
- Using ground shielding when necessary

In short, crosstalk is mainly controlled by reducing electromagnetic coupling between traces through proper **spacing, routing, reference planes and signal separation**.

2.2 Digital Input Circuit Analysis

a) Primary Function

The circuit converts the state of a switch into a stable digital logic signal.

When the switch is closed, the node is connected to $+5\text{ V}$, giving a logic HIGH. When the switch is open, $R_1 = 10\text{ k}\Omega$ pulls the node to ground, giving a defined logic LOW.

Therefore, R_1 is a **pull-down resistor**.

Switch	Input
Open	0 V (LOW)
Closed	5 V (HIGH)

b) Components Inside the Dashed Box

The 50 Hz AC source and the large $R_3 = 1\text{ G}\Omega$ resistance model **weak environmental AC interference**, particularly pickup from mains electricity.

Real wires and PCB traces can couple to nearby mains wiring through small parasitic capacitances. This produces a small 50 Hz noise signal on a high-impedance input.

The $1\text{ G}\Omega$ resistance represents the very weak coupling between the interference source and the signal node.

c) Role of R_1

1) Removing R_1

With R_1 present, the input has a strong path to ground when the switch is open.

The effective resistance to ground is approximately

$$R_{\text{eq}} = R_1 \parallel R_2 \approx 10\text{ k}\Omega$$

Therefore the 50 Hz interference is highly attenuated:

$$\frac{V_{\text{in}}}{V_{\text{noise}}} = \frac{R_{\text{eq}}}{R_3 + R_{\text{eq}}} \approx 10^{-5}$$

If R_1 is removed, only $R_2 = 100\text{ M}\Omega$ provides a weak path to ground:

$$\frac{V_{\text{in}}}{V_{\text{noise}}} = \frac{100\text{ M}\Omega}{1\text{ G}\Omega + 100\text{ M}\Omega} \approx 0.091$$

Hence, removing R_1 increases the noise at the input by roughly

$$\frac{0.091}{10^{-5}} \approx 9,000$$

times.

The simulation should therefore show a much larger 50 Hz oscillation when R_1 is removed.

2) Floating Input

3) Selecting the Value of R_1

The pull-down resistance is a trade-off.

A smaller resistance gives better noise immunity, but draws more current when the switch is closed:

$$I_{R_1} = \frac{V}{R_1}$$

For $R_1 = 10\text{ k}\Omega$,

$$I_{R_1} = \frac{5}{10\,000} = 0.5\text{ mA}$$

and

$$P_{R_1} = \frac{5^2}{10\,000} = 2.5\text{ mW}$$

The resistor must also be low enough that the maximum input leakage current cannot raise the LOW voltage above the digital-input LOW threshold:

$$I_{\text{leak,max}} R_1 < V_{\text{IL,max}}$$

Thus R_1 should be selected considering input leakage, logic thresholds, noise immunity and acceptable current consumption. $10\text{ k}\Omega$ is a common practical value.

d) Arrow and R_2

The arrow most likely represents the signal continuing to the **input of a microcontroller or another digital device**.

$R_2 = 100\text{ M}\Omega$ models the fact that a real input is not perfectly open circuit. It has a very high but finite resistance to ground.

Therefore, R_2 can be called the

Input Resistance (R_{in})

or more generally the **input impedance** of the receiving circuit.

Unlike R_1 , R_2 would normally not be a physical $100\text{ M}\Omega$ resistor placed on the PCB; it is a model of the input itself.

e) Active Signal = 0 V

If the active state should be 0 V, the circuit is reversed:

- connect the resistor from the input to +5 V,
- connect the switch between the input and ground.

The resistor now becomes a **pull-up resistor**.

Switch	Input
Open	5 V (HIGH)
Closed	0 V (LOW)

This is called an

Active-Low Input

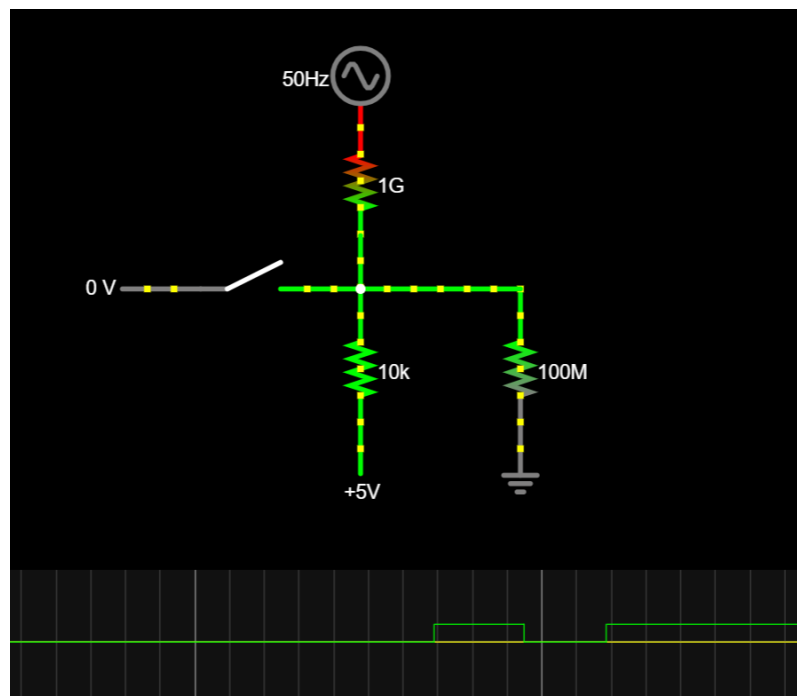


Figure 1: Simulation of the active-low input circuit using a pull-up resistor.

f) Human-Body Interference

The human body can behave like an antenna and couple 50 Hz mains electric fields into an unshielded high-impedance circuit.

Bringing a hand close to the input increases capacitive coupling. Touching it can inject even more noise. If the input is floating, this may cause:

- visible 50 Hz oscillation,
- unstable readings,
- false HIGH/LOW transitions.

2.3 Decoupling and Bulk Capacitors

Decoupling and bulk capacitors both stabilize the power supply, but they handle different types of disturbances.

Decoupling capacitors are small capacitors placed very close to an IC power pin. They supply short, fast current pulses and reduce high-frequency noise.

Typical value:

100 nF

Bulk capacitors are larger capacitors used near the power input, voltage regulator, or high-current loads. They provide more stored energy and reduce slower, larger voltage drops.

Typical values:

10 μF to 100 μF

	Decoupling	Bulk
Purpose	Fast transients, high-frequency noise	Larger/slower voltage drops
Placement	Close to IC power pins	Near power input or regulator
Typical value	100 nF	10–100 μF

In practice, both are usually used together.

2.4 Differential Pairs

A **differential pair** is two PCB traces that carry equal and opposite signals. The receiver detects the voltage difference between the two lines rather than each signal relative to ground.

Differential signalling improves noise immunity because external noise tends to couple similarly into both traces and is rejected by the receiver.

Critical routing rules:

- Route both traces together and keep the spacing between them consistent.
- Keep the two traces as similar in length as required by the interface timing budget.
- Maintain the required differential impedance using the PCB stack-up, trace width and spacing.
- Route over a continuous reference plane and avoid crossing plane gaps.
- Minimize vias and other discontinuities.
- Keep the pair away from noisy switching signals and clocks.
- Avoid unnecessary sharp corners; use smooth or 45° routing.

Common examples include USB, CAN, Ethernet and LVDS.

2.5 Polygon Pours

A **polygon pour** is a large filled copper area on a PCB that is electrically connected to a specific net, most commonly **GND** or a power rail.

It is preferred for power and ground connections because the larger copper area provides:

- lower resistance and voltage drop,
- higher current-carrying capability,
- better heat spreading,
- lower impedance ground return paths,
- improved EMI and signal integrity when used as a ground plane.

For this reason, polygon pours are commonly used instead of narrow traces for high-current power nets and ground connections.

2.6 0–12 V Sensor to 0–5 V ADC Interface

The cheapest solution is a simple **voltage divider** using two resistors:

$$R_1 = 15\text{ k}\Omega, \quad R_2 = 10\text{ k}\Omega$$

where R_1 is connected between the sensor and ADC input, and R_2 between the ADC input and ground.

a) Resistor Selection

The divider output is

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$

At the maximum sensor voltage:

$$V_{out} = 12 \frac{10}{15 + 10} = 4.8\text{ V}$$

which is safely below the ADC's 5 V limit.

The divider power consumption is

$$P = \frac{V^2}{R_1 + R_2} = \frac{12^2}{25\text{ k}\Omega} = 5.76\text{ mW}$$

which satisfies the 6 mW power budget.

Using 1% resistors is recommended to ensure component tolerance does not push the ADC voltage above its limit.

b) Effect of 0.5 mA ADC Leakage

$$V_{TH} = 4.8\text{ V}$$

and

$$R_{TH} = R_1 \parallel R_2 = 15\text{ k}\Omega \parallel 10\text{ k}\Omega = 6\text{ k}\Omega$$

The leakage current produces a voltage error of

$$\Delta V = I_{leak} R_{TH}$$

Therefore,

$$\Delta V = 0.5\text{ mA} \times 6\text{ k}\Omega = 3\text{ V}$$

and the actual ADC voltage at $V_{in} = 12\text{ V}$ becomes

$$\boxed{V_{out} = 4.8 - 3 = 1.8\text{ V}}$$

Hence, the faulty ADC heavily loads the divider and produces a large measurement error. The output is approximately

$$V_{out} = 0.4V_{in} - 3$$

so for inputs below approximately 7.5 V , the ideal calculation would become negative and the real ADC input would clamp near ground, causing significant loss of sensor information.

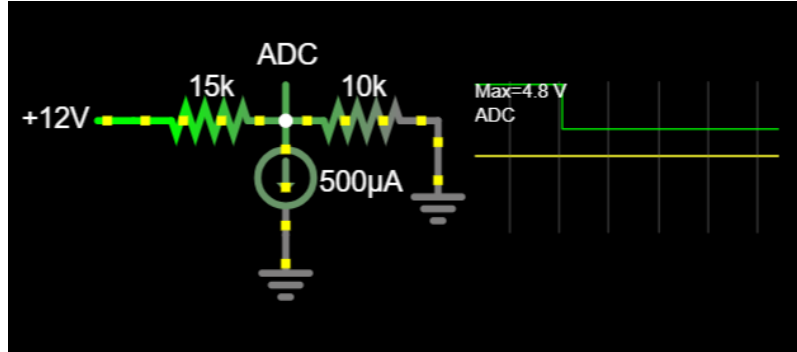


Figure 2: Voltage-divider simulation with 0.5 mA ADC leakage current.

c) Maximum Acceptable Leakage Current

The acceptable leakage depends on the maximum measurement error allowed. Since

$$\Delta V = I_{leak} R_{TH}$$

the general limit is

$$I_{leak,max} = \frac{\Delta V_{max}}{6\text{ k}\Omega}$$

3 Embedded Systems and Architecture

3.1 Simultaneous Interrupt Requests

If two interrupts occur at the same time, the processor services the **higher-priority interrupt first**, while the other remains pending.

In ARM Cortex-M, this is handled by the **NVIC (Nested Vectored Interrupt Controller)**.

- Lower numerical priority value = higher priority.
- A higher-priority interrupt can preempt a lower-priority ISR.
- If both have the same priority, the lower IRQ/exception number is serviced first.

Higher priority executes first; the other stays pending

3.2 Reading Euler Angles from the BNO055

The BNO055 provides Euler angles directly from its internal sensor fusion algorithm. The three angles are stored as signed 16-bit values in six consecutive registers:

Angle	LSB Register	MSB Register
Heading	0x1A	0x1B
Roll	0x1C	0x1D
Pitch	0x1E	0x1F

The microcontroller can read all six bytes starting from register 0x1A. For each angle, the MSB and LSB are combined:

$$Raw = (MSB \ll 8) | LSB$$

and interpreted as a signed 16-bit integer.

When the output unit is degrees:

$$Angle = \frac{Raw}{16}$$

because the BNO055 represents 1° as 16 LSB.

For example:

$$Heading = \frac{(Data[1] \ll 8) | Data[0]}{16}$$

and the same procedure is applied to Roll and Pitch.

Reference: Bosch BNO055 Datasheet, Section 3.6.5.4, page 37 (Euler-angle format), and Sections 4.3.27–4.3.32, pages 66–67 (Euler registers).

3.3 Communication Protocols

Communication protocols are sets of rules that define how embedded devices exchange data. They specify aspects such as data format, timing, addressing, synchronization and error handling.

Common examples include:

- **UART:** Asynchronous serial communication using TX and RX lines. Simple and commonly used for debugging and communication between two devices.
- **I²C:** Synchronous two-wire protocol using SDA and SCL. It supports multiple devices on the same bus using addresses.
- **SPI:** High-speed synchronous protocol using SCLK, MOSI, MISO and CS lines. It is commonly used with sensors, ADCs and memory devices.
- **CAN:** Differential communication bus designed for reliable communication in automotive and industrial systems.

3.4 STM32F4 vs Raspberry Pi Zero vs ATmega

Family	Type	Main Strength	Typical Use Cases
STM32F4	32-bit ARM Cortex-M MCU	High performance, real-time control, many peripherals	Motor control, automotive ECUs, sensors, CAN, control systems
Raspberry Pi Zero	Single-board computer	Runs Linux, high processing capability, networking	High-level processing, data logging, vision, communication gateways
ATmega	8-bit AVR MCU	Simple, low-cost and easy to use	Basic control, simple sensors, small embedded systems

4 Altium PCB Project

5 STM32 Project

GitHub Repository

https://github.com/mrrayyan/LV_Solo_Mission